

Momentum Rebalancing Validation

Calendar timing, trading costs, and reproducible backtests

The monthly-semiannual comparison changes sign across six rebalancing calendars.

Evaluate the claim that more frequent momentum selection improves net portfolio performance. Nine sector ETFs, two signals, six semiannual schedules and four costs form a bounded historical test. The principal finding is calendar sensitivity; inference does not establish a general frequency ranking.

Complete primary calendar grid

Semiannual signal months	0 bps	5 bps	10 bps	25 bps
Jan / Jul	+0.54	+0.37	+0.21	-0.29
Feb / Aug	+0.70	+0.53	+0.36	-0.15
Mar / Sep (reference)	-0.32	-0.49	-0.66	-1.17
Apr / Oct	-0.76	-0.92	-1.09	-1.58
May / Nov	-0.65	-0.82	-0.98	-1.48
Jun / Dec	+0.53	+0.36	+0.19	-0.31

Monthly minus semiannual CAGR, percentage points per year. Positive favors monthly. The 5 bps range is -0.92 to +0.53; March/September is -0.49. All 48 cells across both signals are available in the report. A scenario range is not a confidence interval.

The verification approach

Rules fixed before extension runs with prior history exposure disclosed; closing signals and next-open proxy execution; separate cost accounts; all calendars retained; second implementations reconcile accounting and statistics.

Principal judgment

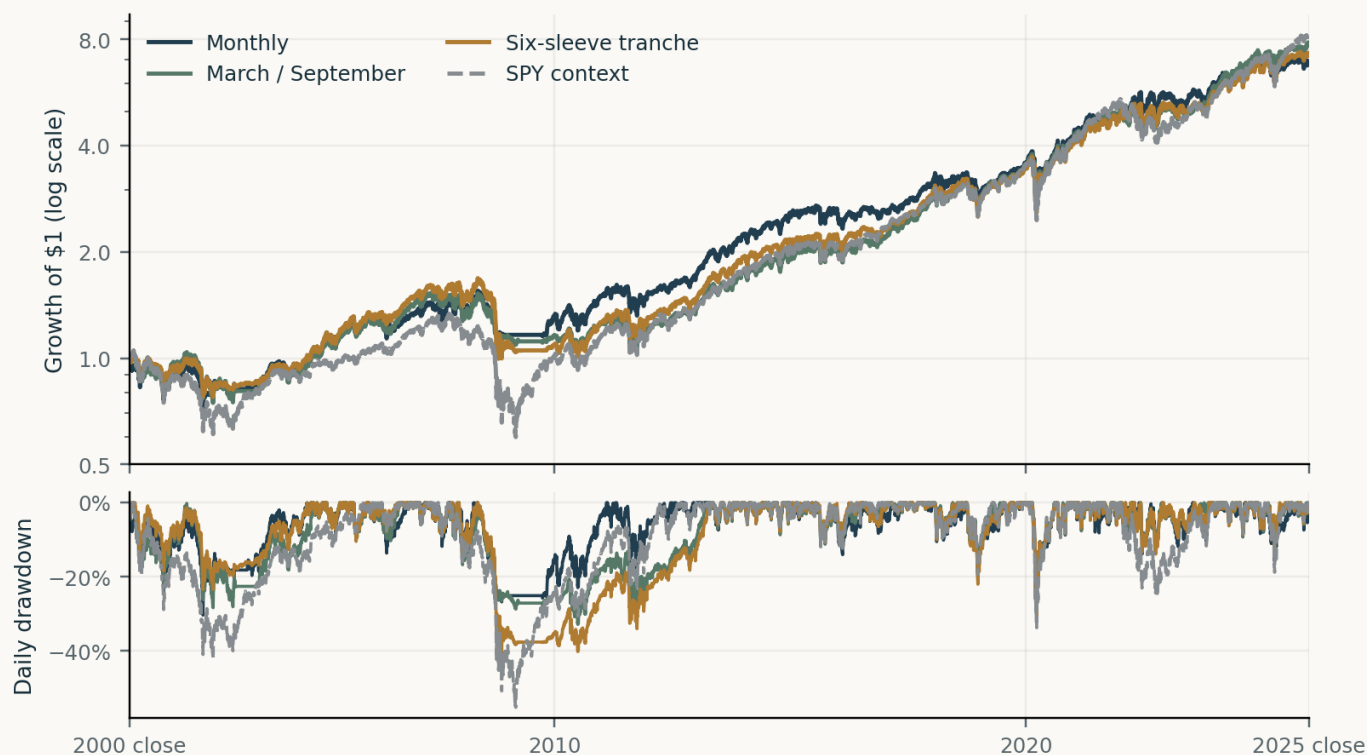
The calculation supports a calendar-sensitive result in this sample. It does not establish an optimal frequency or show that timing luck explains all performance differences. The additional tranche test and uncertainty appear on the next two pages.

Test the Tranche Comparison

Equal initial allocation; autonomous sleeves; no reporting-boundary reset

Six semiannual sleeves each receive one-sixth of initial capital. They first execute on 3 January 2000, then continue without capital transfers or order netting. The 2000 burn-in can create unequal sleeve capital. Reporting from the 2000 year-end close normalizes pooled NAV; it does not re-equalize the sleeves.

Primary wealth and daily drawdown



Each path normalized at the reporting anchor; 12-1 signal, 5 bps per side. SPY is opportunity-cost context with different exposures, not a frequency control. Drawdown uses daily closing NAV.

The mitigation has a tradeoff

Tranche net CAGR is 8.24%, compared with 8.02% monthly and 8.51% for March/September. Its -40.91% maximum drawdown is deeper than both. Pooling calendars does not guarantee better growth or a shallower loss.

Uncertainty and Risk

Monthly risk metrics and paired intervals accompany the return comparison

Primary risk comparison

Portfolio	CAGR	Max DD	Sharpe	TE	Beta	NAV/yr
Monthly	8.02%	-30.32%	0.519	9.97%	0.671	5.04x
Mar / Sep	8.51%	-32.81%	0.557	8.68%	0.718	1.86x
Tranche	8.24%	-40.91%	0.541	7.96%	0.740	1.93x
SPY context	8.76%	-55.19%	0.522	0.00%	1.000	0.00x

Sharpe uses monthly excess returns and captured French RF. TE is annualized monthly tracking error to SPY; beta includes an intercept and uses excess returns. DD is daily maximum drawdown; NAV/yr is gross buys plus sells. Cash earns zero.

Paired 95% intervals for CAGR differences

Monthly minus	Estimate, pp/yr	95% interval, pp/yr
March / September	-0.49	[-3.03, +2.11]
Six-sleeve tranche	-0.22	[-2.44, +2.33]

10,000 paired stationary bootstrap draws of 300 net monthly returns; seed 20261009; primary expected block length 12 months. The same resampled months apply to every account. Compute sampled compounded annual growth, then the paired difference.

Block sensitivity

Mean block	Monthly - Mar/Sep CI	Monthly - tranche CI
6 months	[-2.91, +1.96]	[-2.36, +2.18]
12 months	[-3.03, +2.11]	[-2.44, +2.33]
24 months	[-3.20, +2.18]	[-2.43, +2.38]

Both primary intervals include zero. This does not prove equivalence. The intervals are conditional on realized net returns and resampling assumptions; they do not remove nonstationarity or retrospective design risk. No winning calendar is promoted.

Validation Memo

Calculation reconciled; general frequency superiority unsupported

Scope and disposition

Accept as a reproducible, bounded historical validation case. Compare nine sector ETFs across two signals, six semiannual schedules and four costs, plus six-sleeve pooled accounts. Primary inference is 2001-2025, 12-1, 5 bps. Do not use this case to certify a live strategy or choose an optimal calendar.

Data lineage

Preserved Yahoo responses -> common-session adjusted prices -> portfolio accounts -> accounting replay -> tranche sums -> monthly risk and paired intervals -> public derived CSVs and summary. SHA-256 binds snapshots, configuration and code. French monthly RF is captured separately and joined to 300 months. Hash identity does not certify source accuracy.

Separate implementation checks

Legacy replay: 60 paths. Mechanism replay: 108 paths, 726,732 daily states, maximum relative NAV error $1.78e-14$. Extension: eight tranches, 53,832 daily states and 34,800 monthly rows; capital, cash, units, gross fees, risk and bootstrap percentiles reconcile. The extension verifier imports no project calculation code.

Independence boundary

The same AI agent authored both implementations. Separate algorithms can expose arithmetic and implementation defects but cannot exclude a shared misunderstanding of the specification. No external reviewer agreement, independent price validation or live fill certification is claimed.

Findings and residual risk

The primary difference changes sign across calendars; both paired intervals span zero. Tranching retains a deeper drawdown. Prior history exposure, adjusted-price/opening proxies, fixed costs, zero-interest cash, changing sector exposures and bootstrap nonstationarity remain risks. Written owner result and judgment reviews are accepted; signal-window explanation and observed oral presentation remain pending.

Evidence: Study and downloads | Repository, methods and receipts

Method/source: French Data Library monthly RF; Politis and Romano (1994), The Stationary Bootstrap. Exact URLs and capture hashes appear in the reproduction guide.